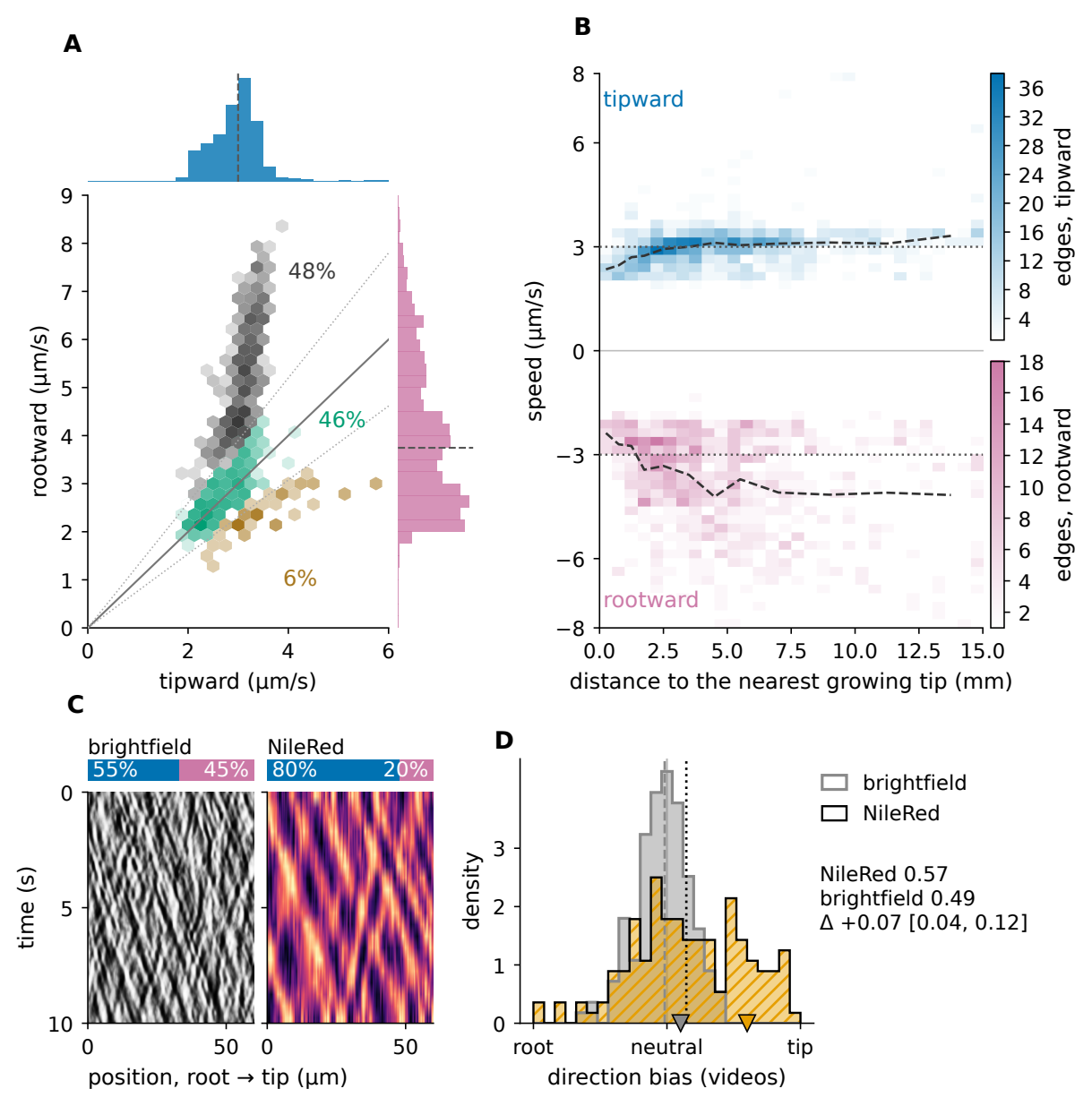

A
B
C
D
48%
46%
6%
rootward (µm/s)
tipward (µm/s)
tipward
rootward
speed (µm/s)
edges, tipward
edges, rootward
distance to the nearest growing tip (mm)
brightfield
NileRed
55%
45%
80%
20%
time (s)
position, root → tip (µm)
density
root
neutral
tip
direction bias (videos)
brightfield
NileRed
NileRed 0.57
brightfield 0.49
Δ +0.07 [0.04, 0.12]